INCH-POUND

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DETAIL SPECIFICATION

CLOTH, TERRY, COTTON, COTTON AND POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE.

1.1 Scope. This specification covers the requirements for a cotton or cotton/polyester terry cloth.

1.2 Classification. The cloth is available in two types.

Type I – 100% Cotton
Type II – 65% Cotton/35% Polyester

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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AMSC N/A FSC 8305

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2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract (see 6.2).

FEDERAL TRADE COMMISSION (FTC)

16 CFR Part 303 - Rules and Regulations under the Textile Fiber Product Identification Act

(Copies are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM20A - Fiber Analysis: Quantitative

(Copies are available online at https://www.aatcc.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D1776/D1776M	- Standard Practice for Conditioning and Testing Textiles
ASTM D3775	- Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M	- Standard Test Methods for Mass Per Unit Area (Weight of Fabric)
ASTM D3990	- Standard Terminology Related to Fabric Defects

(Copies are available online at https://www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.2).

3.2 Basic material. The basic material shall be a woven with a single terry loop pile.

3.3 Yarn. The Type I shall be made of a 100% cotton which has been cleaned, carded, drawn, and spun into singles yarn for the warp pile, warp ground and filling. The Type II shall allow for the ground warp to be made of a blended 65% (±5 percent) cotton and the remainder polyester.

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3.4 Finish. The cloth shall be either bleached or unbleached.

3.5 Non-fibrous material. The total sizing, finishing and other non-fibrous material content shall not exceed 5% when tested as specified in 4.4.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Physical requirements. The finished cloth shall conform to the requirements in Table I.

TABLE I. Physical requirements

Characteristics	Requirement
Fiber Content (Type II) Polyester Cotton	 65% (± 5%) 35% (± 5%)
Weave	Single terry
Yarn ply Warp or filling	 Single ply
Yarns per inch (minimum) Warp Filling	 57 28
Weight, ounces per square yard (minimum)	8.7 oz/sq yd
Finish	Unbleached or bleached treatment

3.8 Width. The minimum acceptable width of the finished cloth, inclusive of selvages, shall be as specified (see 6.2). The selvage shall be 1/4 (± 1/8) inch (6.35 ± 3.18 mm) wide. When a shuttleless loom is utilized the width of the cloth shall be as specified (see 6.2) and shall be excluding the selvages.

3.9 Length. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous pieces, each not less than 40 yards in length.

3.10 Cloth put-up. Each length of cloth furnished shall be put upon rolls, full width with the face side inward.

3.11 Fiber identification label. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.12 Face identification marking. The filling side (back side) of the cloth shall be used as the face side (front side). The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word “Face” at each end of the roll (see 4.4.10.).

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3.13 Marking. The length of each roll shall be legibly marked with indelible ink on a ticket securely attached to the roll.

3.14 Workmanship. The finished cloth shall be uniform in quality and free from irregular defects that can affect usage and/or durability and the defects specified in ASTM D3990 and Section 4. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required, the inspection shall consist of the examinations and tests specified in 4.4.1 through 4.4.12, See Section 3.

4.2.1 First article sample. Unless otherwise specified in the contract (see 6.2), sample unit shall be one continuous five yard length of full width finished cloth.

4.3 Conformance inspection. Conformance testing shall consist of all examinations and tests specified in 4.4.1 through 4.4.12, see Section 3.

4.3.1 Conformance inspection sampling. The sample unit shall be one continuous five-yard length of full width fabric per roll of finished cloth. The number of samples per lot size shall be a specified below.

Lot Size (yards)	Sample Size (rolls)
3200 or less	8
3201 up to and including 10,000	13
10, 001 and over	20

4.3.1.1 In-process inspection. Visual and dimensional examinations of the cloth shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with requirements specified in Section 3. Materials and components, which can be classified, as a defect in accordance with ASTM D3990 for woven cloth and Section 4 shall be removed from production.

4.4 Inspection methods.

4.4.1 Finished cloth tests. The cloth shall be tested as specified in Table II. The finished cloth shall be conditioned as directed in ASTM D1776/D1776M. All test reports shall contain the individual values utilized in expressing the results.

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TABLE II. Finished cloth tests

Characteristics	Requirement	Verification
Fiber analysis	3.7	AATCC TM20A
Non-fibrous	3.7	AATCC TM20A
Weave	3.7	Visual 1/
Yarn ply	3.7	Visual 1/
Yarn per inch	3.7	ASTM D3775
Weight	3.7	ASTM D3776/D3776M, Option C
Finish	3.7	Visual 1/

1/ One 6-inch x 6-inch test specimen shall be used to determine compliance. The results shall be recorded as pass or fail.

4.4.2 Visual examination. The cloth shall be examined on face side only for all defects defined in ASTM D3990 and Section 4 which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit. No linear yard (increment of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45 points. If one roll exceeds the point level per 100 square yards, a second sample of 20 rolls shall be examined only for the individual roll quality. If one or more rolls in the second sample of 20 exceeds the individual point value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

4.4.3.1 Demerit point computation.

$$Points\ per\ 100\ square\ yards\ = \frac{3600 \times T}{W \times Y}$$

Where:
T = total number of points scored in the sample
W = width (in inches) of the cloth purchased
Y = total number of yards inspected

4.4.3.2. Demerit points. Demerit points shall be assigned as follows.

a. For defects up to and including 3 inches in any dimension – one point.
b. For defects exceeding 3 inches but not exceeding 6 inches in any dimension – two points
c. For defects exceeding 6 inches but not exceeding 9 inches in any dimension – three points
d. For defects exceeding 9 inches in any dimension – four points.

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The following defects, when present, shall be scored four points for each yard in which they occur.

e. Baggy, wavy or ridged cloth
f. Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
g. Excessive nippiness

4.4.4 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only as specified in 4.4.1. When the total yardage in the roll does not exceed 100 yards the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards only 100 yards shall be examined. Only one roll shall be taken from any shipping container, unless the number of shipping containers in the lot are less than the number of rolls required for sampling, in which case, all shipping containers shall be sampled.

4.4.5 Width examination. During the yard-by-yard examination, each roll in the sample shall be examined for minimum acceptable width (see 3.8) and selvage width when applicable (see 6.2).

4.4.6 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for gross length. When total yardage in the roll exceeds 100 yards, only 100 yards shall be examined, see 3.9 and 6.2.

4.4.7 Put-up examination. During the yard-by-yard examination, each roll in the sample shall be examined for the cloth's face side being inward, see 3.12.

4.4.8 Fiber identification label examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance to the 16 CFR Part 303 - Rules and Regulations under the Textile Fiber Product Identification Act. The lot shall be unacceptable if two or more rolls in the sample when the fiber identification is missing from either or both ends, two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Product Identification Act see 3.11 and 6.2.

4.4.9 Face identification marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance to the face identification, 3.8. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample are not labeled (see 3.12 and 6.2).

4.4.10 Total yardage in sample examination. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls, see 3.14 and 6.2.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended Use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. National stock number.
c. The specific issue of individual documents referenced (see Section 2).
d. When first article is required (see Section 3.1, Section 4.2).
e. When required, yard-by-yard examination (see Section 4).
f. Name and address of the first article inspection facility; and the name and the address of the Government activity responsible for conducting the first article inspection program.
g. Inspection conditions.
h. Conformance inspection and acceptable quality levels (see Section 4.3)
i. Packaging required (see Section 5.1).

6.3 Information requests. For access to information such as patterns, drawings, standard shade samples of cloth etc. go to Defense Supply Center Philadelphia's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military.

6.4 Subject term (key word) listing.

Rag
Dust cloth
Swipe cloth

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Preparing Activity:
DLA – CT
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